Code No: 135AF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2021
DESIGN AND ANALYSIS OF ALGORITHMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Discuss Strassen's matrix multiplication as well as classical O (n^2) one. Determine when Strassen's method outperforms the classical one.
b) Define Time Complexity. Describe different notations used to represent these complexities. [8+7]

2. List and explain the applications of backtracking. [15]

3. Apply the greedy method to solve Knapsack problem for given instance.
Where n=3, m=20, (p1,p2,p3)=(25,24,15), and weight (w1,w2,w3)=(18,15,10). [15]

4. What is travelling salesman problem? Explain how it can be solved using Branch and Bound method. Show the steps and solution by taking any one example containing atleast 5 cities and 10 ways to travel between them. [15]

5.a) Using branch and bound technique explain the 0/1 knapsack problem.
b) Give a note on FIFO branch and bound solution. [7+8]

6. Explain the Bi-connected components in detail. [15]

7.a) Explain binary search and find its time complexity.
b) Write an algorithm for finding maximum element of an array; perform best, worst and average case complexity with appropriate order notations. [7+8]

8. State and prove the Cook's theorem. [15]

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